

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I S. Subasi (192511296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Subasi

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Scenario-Based Question

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CO2 Assessment

cloud computing

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1. cloud Architecture for a High-Demand video streaming platform.

Scenario.

A media company wants to migrate its video streaming platform to the cloud. The platform must support millions of concurrent users while providing smooth streaming with minimal buffering.

Proposed cloud Architecture.

highly scalable, distributed cloud architecture. using object storage, CDN, load balancing, auto-scaling and caching.

Architecture:

users → DNS / Global Traffic Manager - CDN → Load Balancer → Auto scaling Application servers - video / Object storage.

1. cloud object storage.

All video content should be stored in high durable cloud object storage. video should be encoded into multiple resolutions and delivered using adaptive streaming technologies such as HLS or MPEG-DASH.

2. Content delivery Network (CDN)

A CDN should be used to cache popular video content at geographically distributed edge locations.

When a user requests a video, the CDN delivers it from a nearby edge location, instead of repeatedly accessing the origin storage.

Benefits:-

- * Reduce Latency
- * Minimize buffering
- * Reduce origin server load
- * Improve global content delivery.

3. Load Balancing

A cloud Load Balancer distributes incoming

Application requests across multiple healthy servers.

This prevents a single server from becoming a bottleneck and improves application availability.

4. Auto-scaling

Auto-scaling automatically adjusts computing resources according to demand.

- * During peak hours, additional instances or containers are launched.

- * During low-demand periods, unnecessary resources are removed.

5. Caching strategy

multiple caching layers can be implemented:

- * CDN cache: stores frequently accessed

video content.

- * Application cache: stores frequently accessed requested metadata.

- * Database cache: Reduces repeated database queries.

code optimization strategies

1. Use CDN caching to reduce repeated data transfer from origin servers.

2. Implement auto-scaling to avoid paying for unused computing resources.

3. Use storage lifecycle policies to move older content to lower-cost storage tiers.

22 Fintech startup - container orchestration and multi-cloud.

A fintech startup can use Kubernetes to manage containerized applications consistently across multiple cloud providers. Applications should be packaged as standard container images and Deployed using CI/CD pipelines and Infrastructure as Code (Iac).

* Service Discovery: Kubernetes services and DNS allow containers to communicate using stable service names.

* Scaling :- Horizontal pod Autoscaler automatically increases or decreases containers based on workload.

* Multi-cloud Benefits :- High Availability, reduced vendor lock-in, disaster recovery and flexibility.

* Risks : Increased complexity, higher management costs, data synchronization issues, security challenges and cross-cloud network costs.

Conclusion:-

kubernetes with CI/CD, service discovery and auto-scaling provides a consistent and scalable multi-cloud deployment strategy for k8s applications.

3. E-Learning platform - Backup and Disaster recovery.

The e-learning platform should use automated cloud Backups, Data Replications and Versioning to prevent Data loss.

Backup: Schedule regular Database and file Backups.

Versioning: Enable Object versioning to recover accidentally Deleted or Modified files.

Replication: Replicate Critical Data to a Different Cloud region for Disaster Recovery.

RTO: Use automated Recovery and failover to restore services quickly.

RPO: Use frequent Backups and continuous

Replication to minimize data loss

Business continuity: Maintain Backup copies,

Monitor Backup status, test recovery

procedures regularly, and use a secondary

region during failure

Conclusion:-

A combination of Backup versioning, Replication, and automated recovery ensures data protection and continuous operation during failure.

Government Agency - Hybrid cloud.

The Government Agency can use a Hybrid cloud Architecture, keeping sensitive applications and data on-premises while hosting non-sensitive Applications in the public cloud.

* Secure Communication:
use VPN or Dedicated private connection with firewalls and Network segmentation.

* Identity Management:
Implement centralized IAM with MFA, RBAC, SSO and Least-privilege access.

* Encryption:
Encrypt sensitive data at rest and in transit using secure encryption and key management.

* Monitoring: -
use centralized logging, Security Monitoring and regular audits.

* challenges :-
complex management, data synchronization,
security integration, compliance requirements
and higher operational costs.

conclusion:

Hybrid cloud provides flexibility and
scalability while keeping sensitive
government data secure and
compliant.

5. SaaS provider - Multi-Region High Availability
The SaaS provider should deploy its
application across multiple cloud regions
to ensure continuous availability even if
one region fails.

* Multi-Region Deployment:

Deploy application servers
and databases in at least
two separate regions.

2. Data Replication:-

Replicate data between regions using synchronous or asynchronous replication based on RPO and consistency requirements.

3. Failover:-

Use a global load balancer or Traffic manager to automatically redirect users to a healthy region when one region fails.

4. Auto-scaling:-

Automatically increases resources in the active region fails during high traffic or failover situation.

4. Monitoring:-

Continuously Monitor CPU, memory, latency, errors, database health, and replication status

Alerting :-

Send immediate alerts for regional failure, high error rates and replication issues.

Conclusion =

A Multi-region architecture with data replication, automated failover, auto-scaling and continuous monitoring ensure high availability and fault tolerance for the SaaS application.